LIM-4.A.1 When the ratio of two functions tends to 0/0 or ∞/∞ in the limit, such forms are said to be indeterminate.

LIM-4.A.2 Limits of the indeterminate forms 0/0 or ∞/∞ may be evaluated using L'Hospital's Rule.

4-7 L'Hospital's Rule

Get Ready...

Evaluate the following limit: $\lim_{x\to 4} \frac{x^2+2x-24}{x^2-16}$

$$= \lim_{x\to 4} \frac{(x+6)(x-4)}{(x-4)(x+4)}$$

$$= \lim_{x\to 4} \frac{x+6}{x+4} = \frac{10}{8} = \frac{5}{4}$$

When working with a limit, our first approach is to attempt direct substitution. However, direct substitution often leads to an **indeterminant form**.

Indeterminant Form	If a limit of a function tends to $\frac{0}{0}$ or $\frac{\infty}{\infty}$, this is considered to be of **indeterminate form**. Indeterminate form simply implies that the limit cannot be determined without additional analysis. **Note:** There are other indeterminate forms including $0 \cdot \infty, \infty - \infty, 1^{\infty}$, and 0^0. However, these forms will not appear on the AP Exam.

In Unit 1, we evaluated limits like the warm-up above by using algebraic manipulation: factoring and reducing. However, this method of evaluating limits of indeterminate form is very limited and does not work with many types of limits.

For example, consider the following limit: $\lim_{x \to 0} \frac{e^{2x}+x-1}{x^2+4x}$ $\quad \frac{1+0-1}{0+0} = \frac{0}{0}$

Direct substitution leads to the indeterminate form: $\frac{0}{0}$. However, none of the methods we used in Unit 1 will help us evaluate the limit.

Thankfully, there is another method that will help us evaluate limits of indeterminate forms $\frac{0}{0}$ and $\frac{\infty}{\infty}$.

L'Hospital's Rule (Theorem)	Let f and g be functions that are differentiable on an open interval (a, b) containing c, except possibly at c itself. Assume $g'(x) \neq 0$ for all x in (a, b), except possibly at c itself. If $\lim_{x \to c} f(x) = 0$ and $\lim_{x \to c} g(x) = 0$ or $\lim_{x \to c} f(x) = \pm\infty$ and $\lim_{x \to c} g(x) = \pm\infty$ then $\lim_{x \to c} \frac{f(x)}{g(x)} = \lim_{x \to c} \frac{f'(x)}{g'(x)}$

Important notes for L'Hospital's Rule
A. This is a **THEOREM**. Theorems require that we always check conditions BEFORE applying the conclusion. B. $\frac{0}{0}$ and $\frac{\infty}{\infty}$ are indeterminate forms and NOT numbers. Do not write $= \frac{0}{0}$ or $= \frac{\infty}{\infty}$ on your paper! C. When working with a limit question, follow the order… 1. Write the limit of the numerator and the limit of the denominator separately. 2. If BOTH limits approach 0 or ∞, apply L' Hospital's Rule. 3. After applying L' Hospital's Rule, find the limit of the numerator and denominator separately (again). 4. Once the limit of the numerator and denominator do not create an indeterminate form, the ratio of these limits is the answer.

Example 1: $\lim_{x \to 1} \frac{\ln x}{x-1}$ $\frac{0}{0}$

$\lim_{x \to 1} (\ln x) = 0$ and $\lim_{x \to 1} (x-1) = 0$

By L'Hospital's Rule:

$$\lim_{x \to 1} \frac{\ln x}{x-1} = \lim_{x \to 1} \frac{\frac{1}{x}}{1} = \frac{1}{1} = \boxed{1}$$

Example 2: $\lim_{x\to 0} \frac{e^{2x}+x-1}{x^2+4x}$

$\lim_{x\to 0}(e^{2x}+x-1)=0$ and $\lim_{x\to 0}(x^2+4x)=0$

By L'Hospital's Rule:

$$\lim_{x\to 0}\frac{e^{2x}+x-1}{x^2+4x} = \lim_{x\to 0}\frac{2e^{2x}+1}{2x+4} = \frac{2+1}{0+4}$$

$$= \frac{3}{4}$$

Example 3: $\lim_{x\to\infty} \frac{\ln x}{\sqrt[3]{x}}$

$\lim_{x\to\infty} (\ln x) = \infty$ and $\lim_{x\to\infty} (\sqrt[3]{x}) = \infty$

By L'Hospital's Rule:

$$\lim_{x\to\infty} \frac{\ln x}{\sqrt[3]{x}} = \lim_{x\to\infty} \frac{\frac{1}{x}}{\frac{1}{3x^{2/3}}}$$

$$= \lim_{x\to\infty} \frac{3}{x^{1/3}}$$

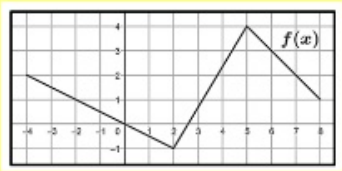
$$= 0$$

$\sqrt[3]{x} = x^{1/3}$

$\frac{1}{3}x^{-2/3} = \frac{1}{3x^{2/3}}$

$\frac{1}{x} \cdot \frac{3x^{2/3}}{1}$

$= \frac{3x^{2/3}}{x}$



x	-2	0	4	6
$g(x)$	-4	0	-3	7
$g'(x)$	4	6	-1	2
$g''(x)$	3	2	0	-5

Example 4: $\lim\limits_{x\to 4} \frac{g(x)+3e^{2x-8}}{4-x}$

$$\lim_{x\to 4}\left(g(x)+3e^{2x-8}\right) = g(4)+3e^{0} = -3+3 = 0$$

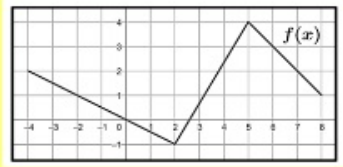
$$\lim_{x\to 4}(4-x) = 4-4 = 0$$

By L'Hospital's Rule:

$$\lim_{x\to 4}\frac{g(x)+3e^{2x-8}}{4-x} = \lim_{x\to 4}\frac{g'(x)+3e^{2x-8}(2)}{-1}$$

$$= \frac{g'(4)+6e^{0}}{-1} = \frac{-1+6}{-1}$$

$$= \boxed{-5}$$



x	-2	0	4	6
$g(x)$	-4	0	-3	7
$g'(x)$	4	6	-1	2
$g''(x)$	3	2	0	-5

Example 5: $\lim\limits_{x\to 0} \frac{g'(x)-6\cos(4x)}{f(g(x))}$

$$\lim_{x\to 0}\left(g'(x)-6\cos(4x)\right) = g'(0)-6\cos 0 = 6-6=0$$

$$\lim_{x\to 0}\left(f(g(x))\right) = f(g(0)) = f(0) = 0$$

By L'Hospital's Rule:

$$\lim_{x\to 0}\frac{g'(x)-6\cos(4x)}{f(g(x))} = \lim_{x\to 0}\frac{g''(x)+6\sin(4x)\cdot 4}{f'(g(x))\cdot g'(x)}$$

$$= \frac{g''(0)+24\sin 0}{f'(g(0))\cdot g'(0)}$$

$$= \frac{2+0}{f'(0)\cdot g'(0)} = \frac{2}{-\frac{1}{2}6} = \frac{2}{-3}$$